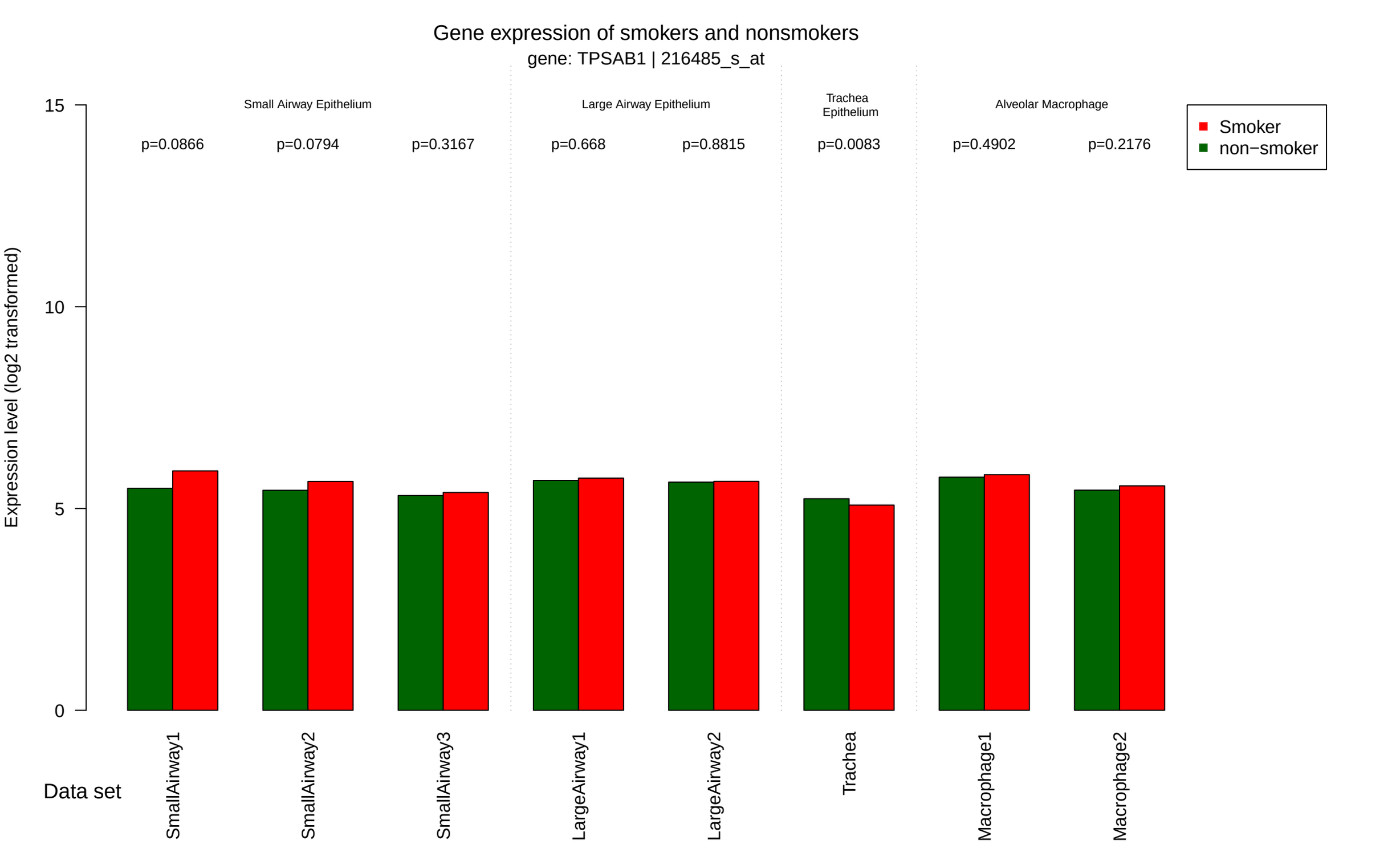
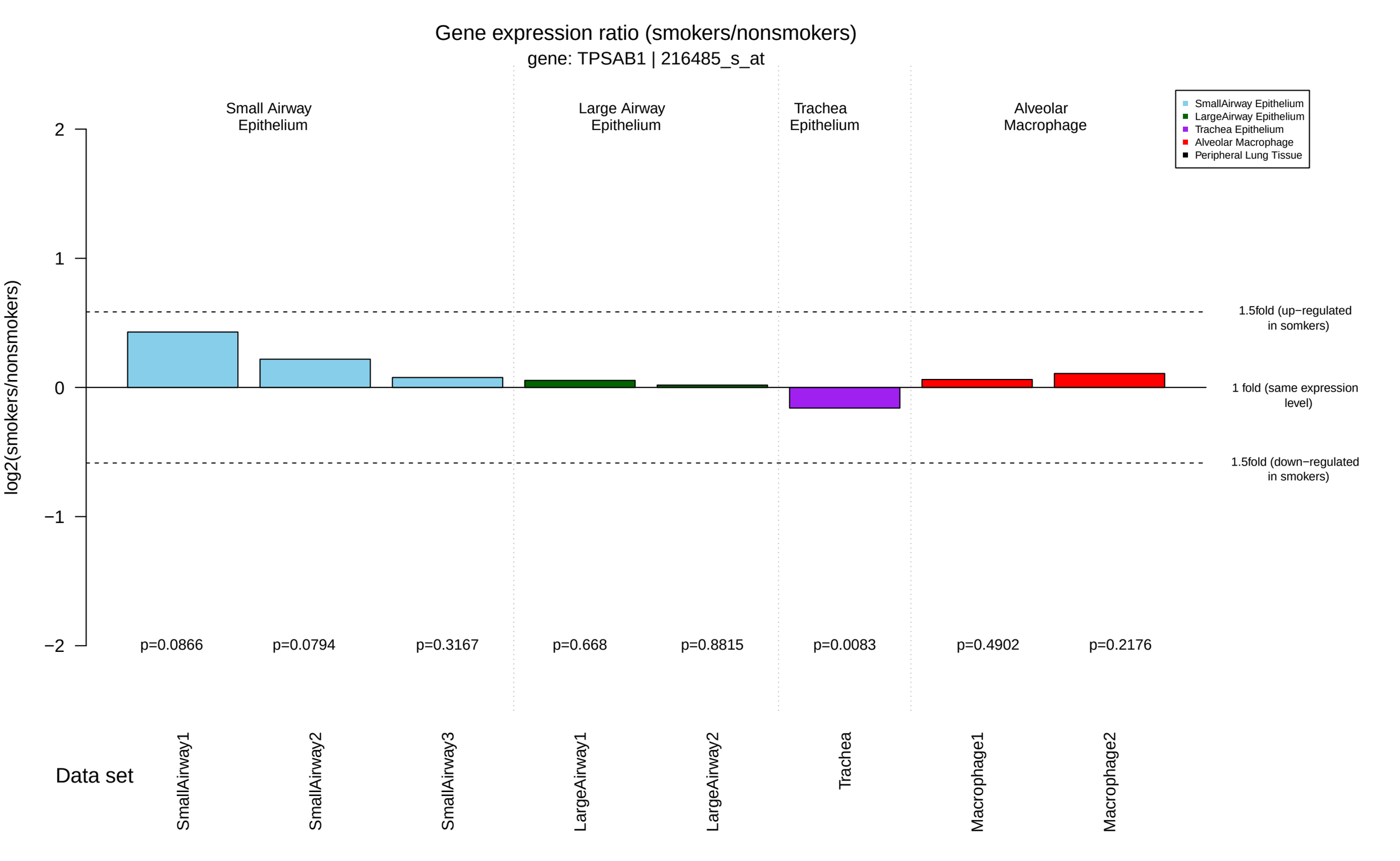
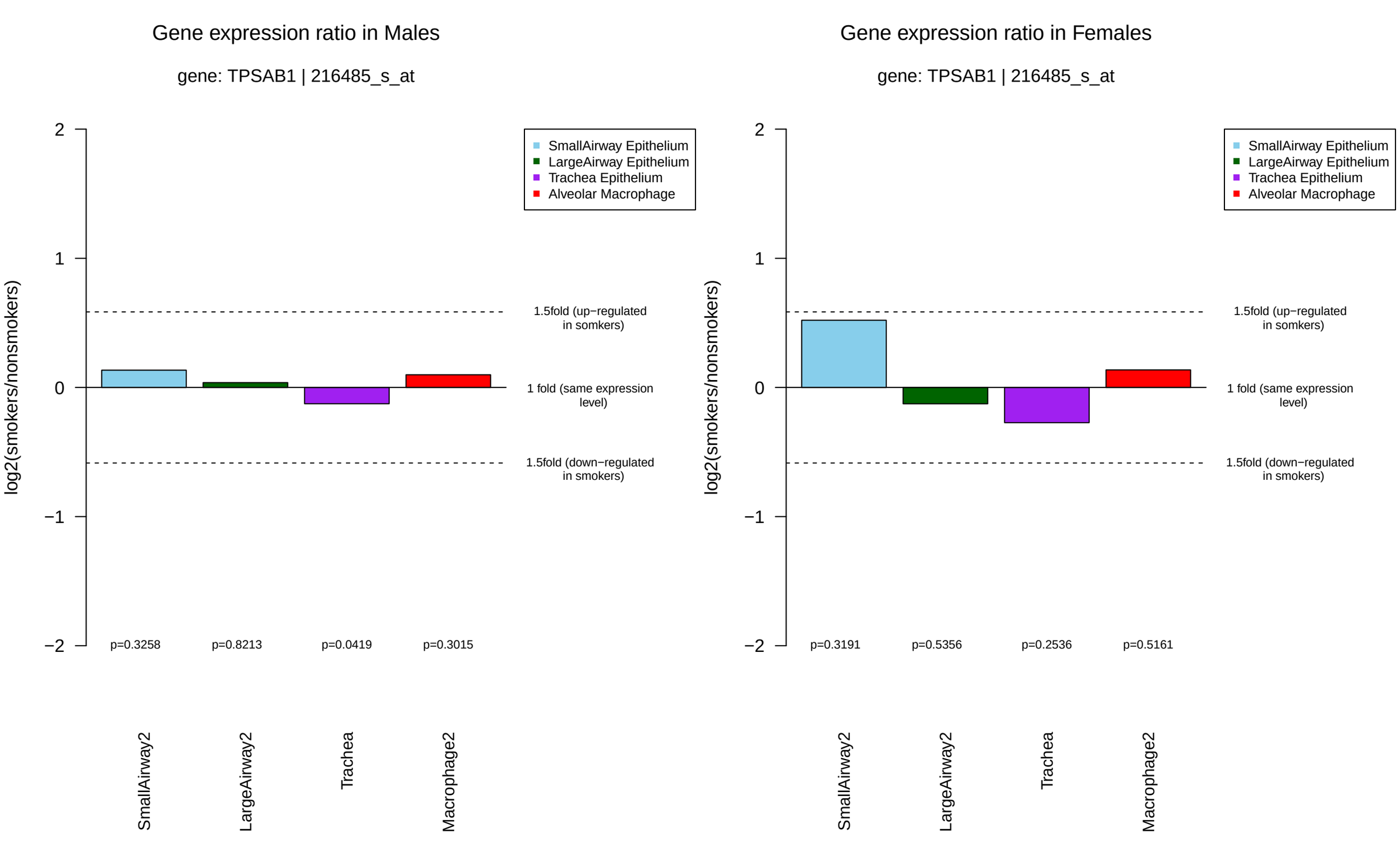
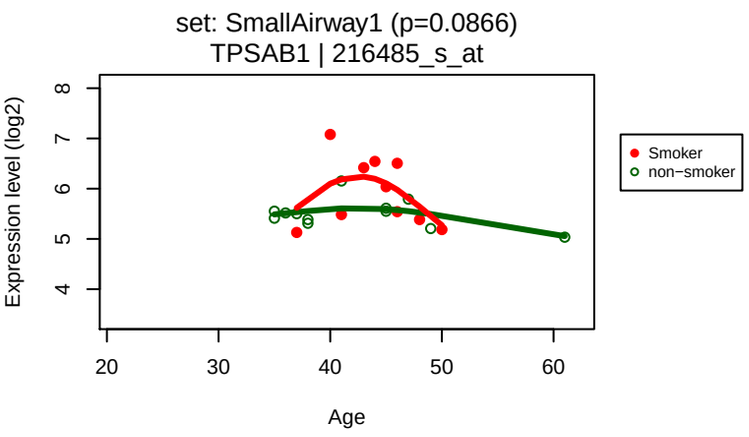
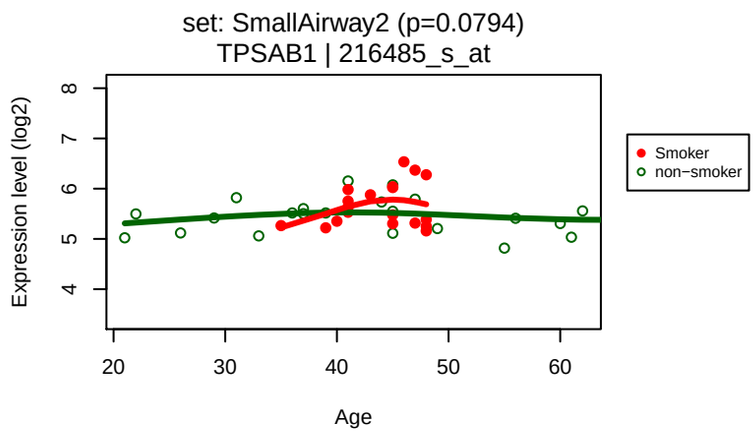
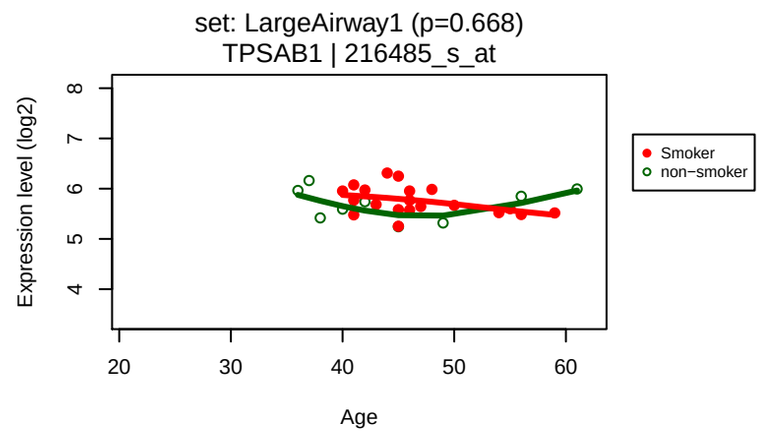

Gene expression of smokers and nonsmokers
gene: TPSAB1 | 216485_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0866
p=0.0794
p=0.3167
p=0.668
p=0.8815
p=0.0083
p=0.4902
p=0.2176
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

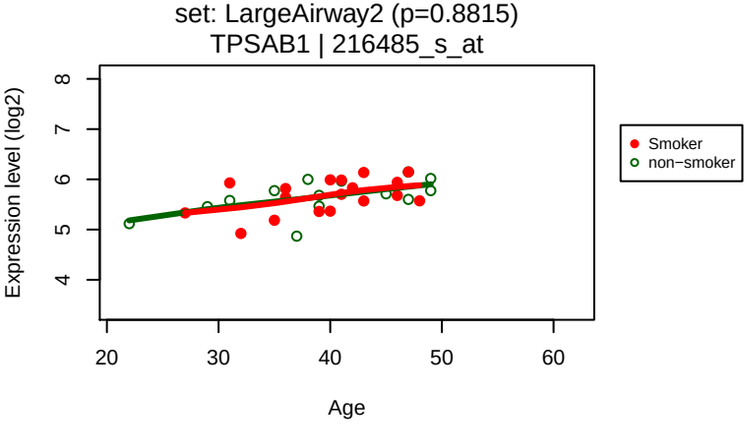

Gene expression ratio (smokers/nonsmokers)
gene: TPSAB1 | 216485_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0866
p=0.0794
p=0.3167
p=0.668
p=0.8815
p=0.0083
p=0.4902
p=0.2176
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

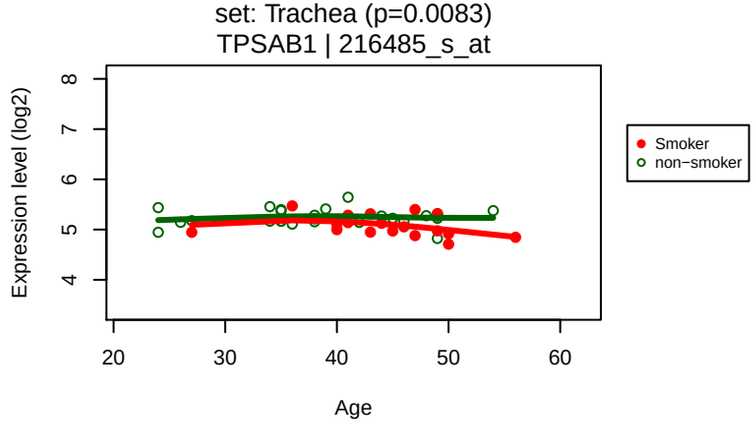

Gene expression ratio in Males
gene: TPSAB1 | 216485_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3258
p=0.8213
p=0.0419
p=0.3015
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TPSAB1 | 216485_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3191
p=0.5356
p=0.2536
p=0.5161
SmallAirway2
LargeAirway2
Trachea
Macrophage2

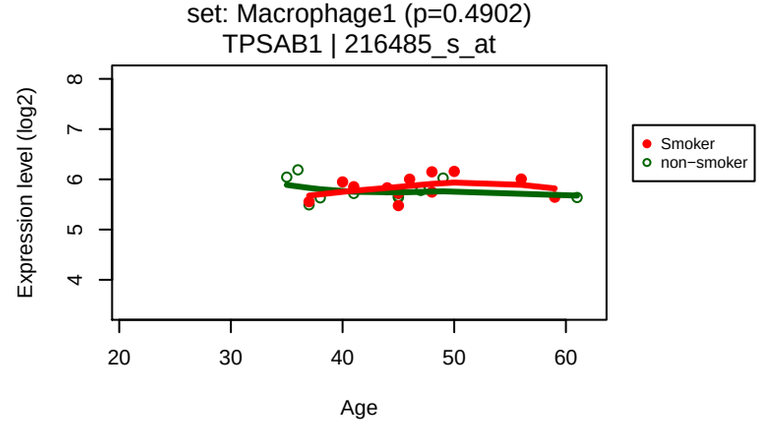

set: SmallAirway1 (p=0.0866)
TPSAB1 | 216485_s_at
Expression level (log2)
Age
Smoker
non−smoker

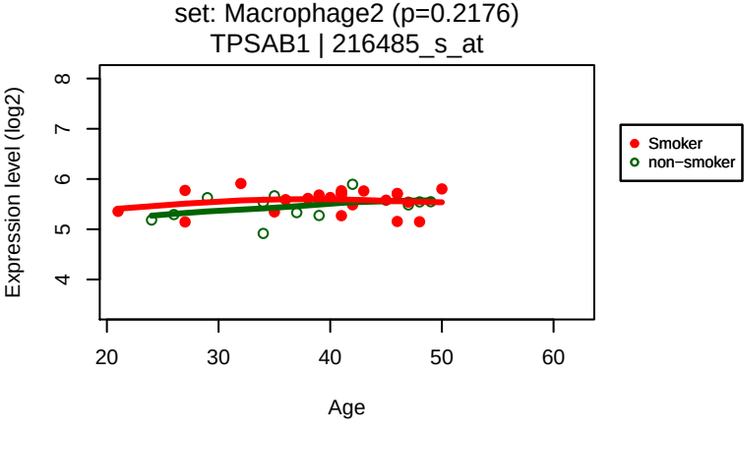

set: SmallAirway2 (p=0.0794)
TPSAB1 | 216485_s_at
Expression level (log2)
Age
Smoker
non−smoker

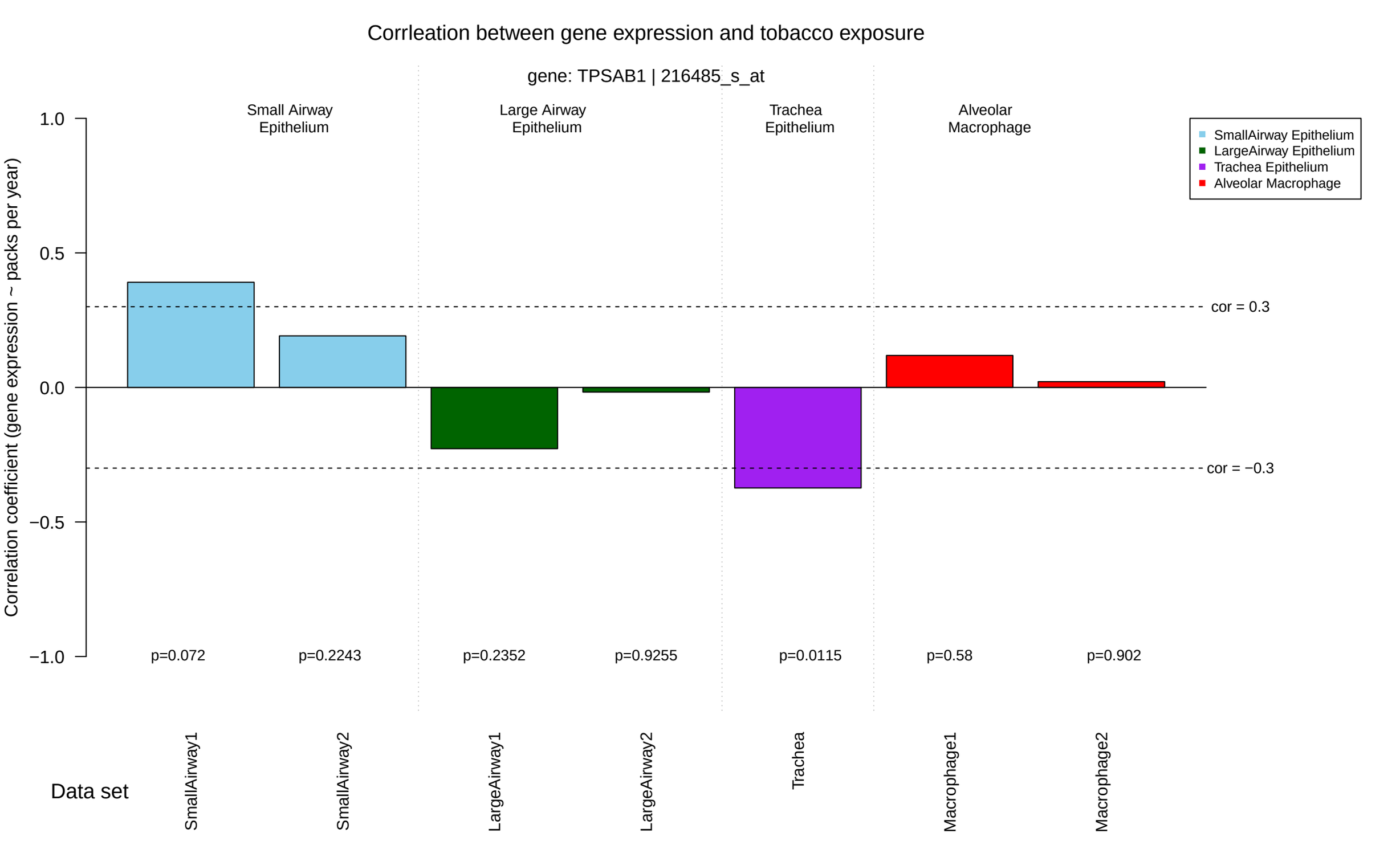

set: LargeAirway1 (p=0.668)
TPSAB1 | 216485_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8815)
TPSAB1 | 216485_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0083)
TPSAB1 | 216485_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4902)
TPSAB1 | 216485_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2176)
TPSAB1 | 216485_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TPSAB1 | 216485_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.072
p=0.2243
p=0.2352
p=0.9255
p=0.0115
p=0.58
p=0.902
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2